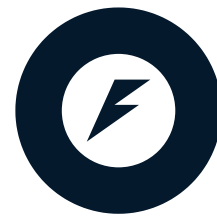


PUT and DELETE operations

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PUT vs. DELETE

PUT Operations

- Traditional use: update an existing object
- Parameters sent via query string as well as request body
- Requires an application or framework
 - e.g. `cURL` , `requests`

```
api = "http://moviereviews.co/reviews/1"
body = {"text": "A fantastic movie!"}
response = requests.put(api, json=body)
```

DELETE Operations

- Traditional use: delete an existing object
- Parameters sent via query string as well as request body
- Requires an application or framework
 - e.g. `cURL` , `requests`

```
api = "http://moviereviews.co/reviews/1"
response = requests.delete(api)
```

Referencing Existing Objects

- No ORM, so app must map object to ID
- Database ID - unique identifier
- `_id` convention for database IDs
 - `review_id`: Table `reviews`, column `id`
 - Same convention in frameworks with ORM

```
from pydantic import BaseModel

class DbReview(BaseModel):
    movie: str
    num_stars: int
    text: str
    # Reference database ID of Reviews
    review_id: int
```

Handling a PUT Operation

PUT endpoint to update an existing movie review:

- Endpoint: `/reviews`
- Input: `DbReview` (from previous slide)
- Output: `DbReview`

```
@app.put("/reviews", response_model=DbReview)
def update_review(review: DbReview):
    # Update the movie review in the database
    db_review = crud.update_review(review)
    # Return the updated review
    return db_review
```

Handling a DELETE Operation

DELETE endpoint to remove an existing movie review:

- Endpoint: `/reviews`
- Input: `DbReview`
- Output: `{}`

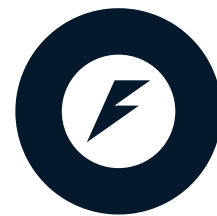
```
@app.delete("/reviews")
def delete_review(review: DbReview):
    # Delete the movie review from the database
    crud.delete_review(review)
    # Return nothing since the data is gone
    return {}
```

Let's practice!

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Handling errors

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Two Main Reasons To Handle Errors

User error

- Invalid or outdated URI
- Missing or incorrect input

```
@app.delete("/items")
def delete_item(item: Item):
    if item.id not in item_ids:
        # Return an error
    else:
        crud.delete_item(item)
    return {}
```

Server error

- Something else happened

```
@app.delete("/items")
def delete_item(item: Item):
    try:
        crud.delete_item(item)
    except Exception:
        # Return an error
    return {}
```


HTTP Status Codes: "Levels of Yelling"

- Enables API to provide status in response
 - Success, failure, error, etc.
 - Specific codes defined in HTTP protocol
 - Range: 100 - 599
 - Categorize by first number (1 - 5)
1. Informational responses (100 - 199)
 2. Successful responses (200 - 299)
 3. Redirection messages (300 - 399)
 4. Client error responses (400 - 499)
 5. Server error responses (500 - 599)

¹ <https://developer.mozilla.org/en-US/docs/Web/HTTP/Status>

Common HTTP Status Codes

Success (200 - 299)

- 200 OK
 - Default success response
- 201 Created
 - Specific to POST operation
- 202 Accepted
 - Noncommittal. "Working on it"
- 204 No Content
 - Success! Nothing more to say

Other responses

- 301 Moved Permanently
 - URI changed permanently
- 400 Bad Request
 - Client error
- 404 Not Found
 - Server cannot find the requested resource
- 500 Internal Server Error
 - Server has encountered a situation it does not know how to handle

Handling Errors With Status Codes

```
from fastapi import FastAPI, HTTPException

app = FastAPI()

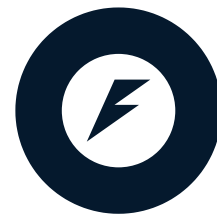
@app.delete("/items")
def delete_item(item: Item):
    if item.id not in item_ids:
        # Send response with status 404 and specific error message
        raise HTTPException(status_code=404, detail="Item not found.")
    else:
        delete_item_in_database(item)
        return {}
```

Let's practice!

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Using async for concurrent work

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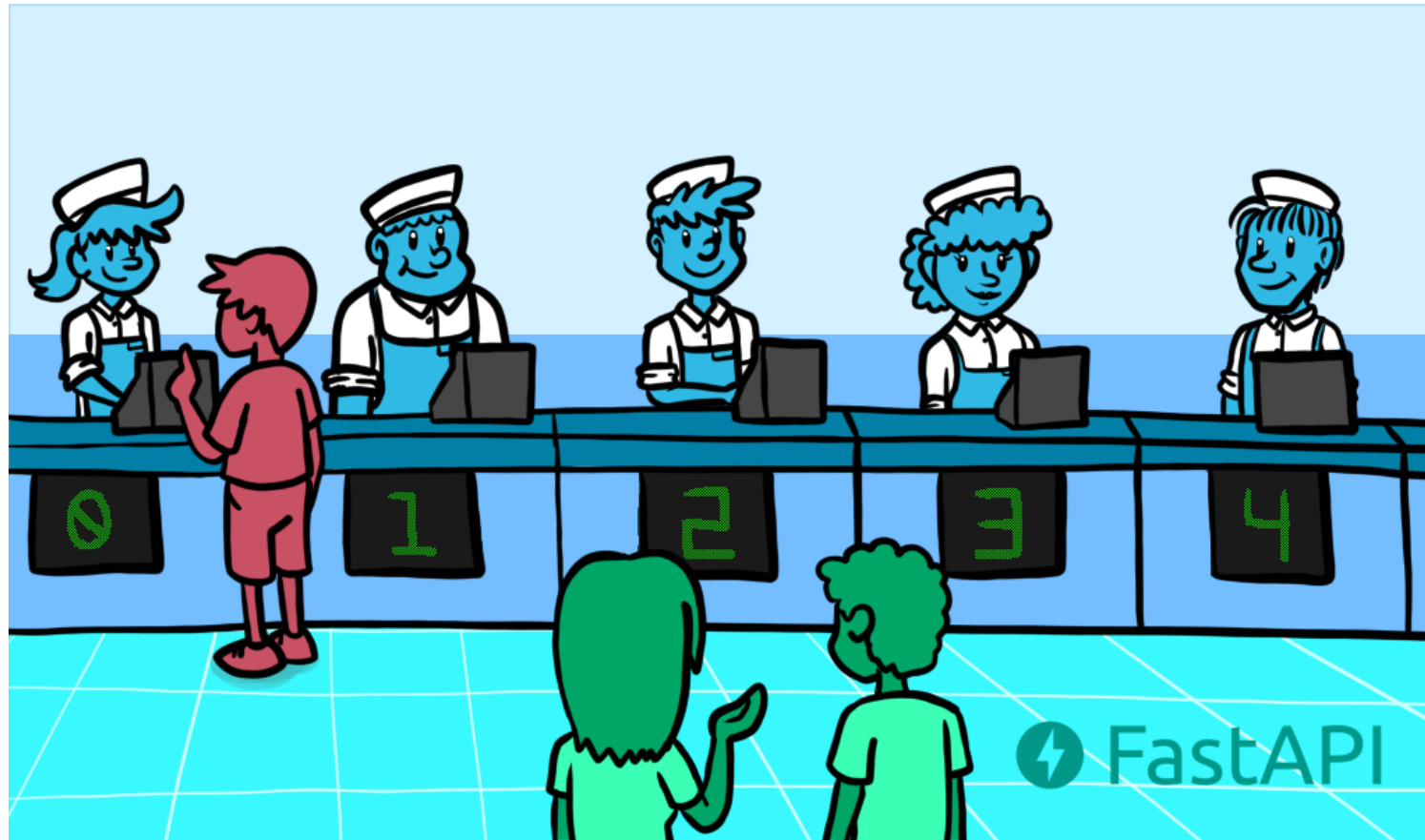


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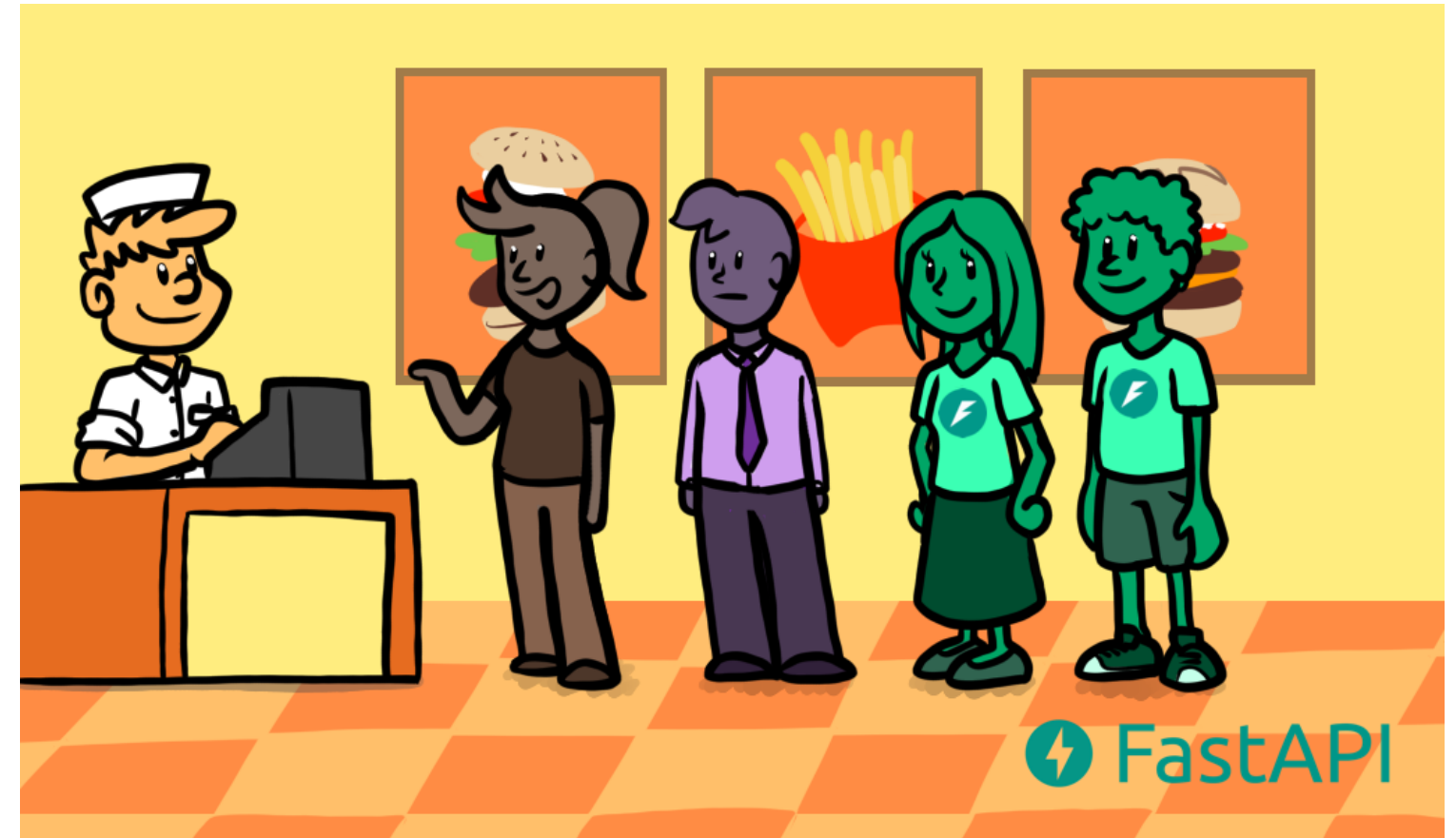
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Why use async? Concurrent Burgers!

Sequential Burgers



Concurrent Burgers



¹ <https://fastapi.tiangolo.com/async/>

async in practice

Sequential Burgers

Defining a function to get burgers

```
# This is not asynchronous
def get_sequential_burgers(number: int):
    # Do some sequential stuff
    return burgers
```

Calling the function sequentially

```
burgers = get_burgers(2)
```

Concurrent Burgers

Defining a function to get burgers

```
async def get_burgers(number: int):
    # Do some asynchronous stuff
    return burgers
```

Calling the function asynchronously

```
burgers = await get_burgers(2)
```


FastAPI with async

If we can:

```
results = await some_library()
```

Then use `async def` :

```
@app.get('/')
async def read_results():
    results = await some_library()
    return results
```

Note Only use `await` inside of functions created with `async def`

When to use async

Use async

If our application needs to wait for other systems to respond

- External API
- Database

Examples

- HTTP requests
- Querying databases
- Reading files

Don't use async

CPU-heavy tasks

Examples

- Audio or image processing
- Computer vision
- Machine Learning
- Deep Learning

If we aren't sure

Let's practice!

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